



Rethinking College Apps: Gamification in College ERP Systems

Enhancing Student Engagement Through Game Mechanics

Members - Krishna Ghodke, Tanishka Jadhav, Kaustubh Wankhede

Introduction





Introduction

What is Gamification?

Definition: Applying game-like elements in non-game contexts.

Example:  Duolingo's streaks, XP points, leaderboards.

Why ERP Systems Need Gamification?

 Students find traditional ERP systems boring and rarely engage with them.

 Gamification makes tasks (attendance , assignment submissions ) more interactive.



Research Objective



Research Objective

Goal:  Increase student interaction with ERP by using gamification.

Core Features:

 XP & Badges for attendance , assignments , and participation .

 Leaderboards  to boost motivation.

 Streaks  to encourage consistency.



Background & Problem Statement



Background & Problem Statement



Current Issues in ERP Usage



Low engagement rates – students rarely use the ERP beyond necessities.



ERP feels like a task manager, not an interactive tool.



How Gamification Can Help?



Enhances motivation through rewards & competition.



Encourages frequent interactions by making tasks more engaging.



Methodology



Methodology



Research Methods



Literature review on gamification in education.



Surveyed students to assess ERP engagement levels.



Studied existing gamified platforms like Duolingo 🏆 & Khan Academy 🎓.



Implementation Approach



Designed mockups for a gamified ERP UI.



Identified key engagement triggers to enhance user interaction.

Literature Review



Literature Review - Key Findings from the Research

 **Enhances Engagement & Motivation** – Studies (Hamari et al., 2014) show gamification increases student participation.

 **Improves Learning Outcomes** – Gamified environments boost retention and understanding (Deterding et al., 2011).

 **Encourages Consistency** – Features like streaks and rewards (e.g., Duolingo) promote continuous learning (Zichermann & Cunningham, 2011).

 **Challenges Exist** – Over-reliance on rewards may reduce intrinsic motivation (Deci et al., 1999); competition must be balanced (Sailer et al., 2017).

 **Best Practices** – Personalization, adaptive difficulty, and meaningful rewards improve long-term effectiveness (Kapp, 2012).



Duolingo Case Study



Case Study: Duolingo's Gamification Strategy and Its Impact on Learning Retention

1. Introduction

- Duolingo is a globally recognized language-learning platform that has redefined digital education through gamification.
- It has over **500 million** users, making it one of the most popular language-learning tools worldwide.
- The success of Duolingo is largely attributed to its game-like structure, which makes learning engaging, interactive, and habit-forming.
- This case study examines **Duolingo's gamification strategies**, their psychological basis, and their impact on **user engagement and retention**.

2. Duolingo's Gamification Features

Duolingo incorporates various game mechanics to enhance the learning experience:

- **XP (Experience Points):**
 - Users earn XP for completing lessons, similar to leveling up in a video game.
 - This feature rewards consistency and provides an immediate sense of progress.
- **Streaks:**
 - Users are encouraged to practice daily, as streaks track the number of consecutive days they complete lessons.
 - This fosters habit formation and motivates users to maintain their progress.
- **Leaderboards:**
 - Users compete in leagues based on XP earned, creating a **healthy sense of competition**.
 - This social aspect encourages users to stay engaged and advance in rankings.
- **Badges & Achievements:**
 - Users unlock **milestones and achievements**, which provide additional motivation and recognition.
 - Achievements serve as psychological rewards, reinforcing learning behavior.
- **Hearts System:**
 - A 'lives' system limits the number of mistakes a learner can make in a session.
 - This encourages careful learning and reduces the likelihood of careless errors.
- **Gamified Challenges & Quests:**

- Periodic **special challenges** and **limited-time events** provide additional motivation for deeper engagement.
- These challenges introduce variety, preventing monotony in the learning process.

3. Psychological Foundations of Duolingo's Gamification

Duolingo's approach is based on well-established psychological theories:

- **Self-Determination Theory (Deci & Ryan, 1985):**
 - **Autonomy:** Users can control their learning pace and choose lessons freely.
 - **Competence:** Progressive difficulty levels ensure learners feel challenged but not overwhelmed.
 - **Relatedness:** Leaderboards and social features help users connect with others, increasing motivation.
- **Operant Conditioning (Skinner, 1953):**
 - Rewards (XP, badges) reinforce positive behavior (consistent learning).
 - Penalties (losing streaks, hearts) discourage negative behavior (inconsistent learning or careless mistakes).
- **Flow Theory (Csikszentmihalyi, 1990):**
 - The balance between challenge and skill level prevents frustration or boredom, ensuring a **state of flow** where users remain fully immersed in learning.

4. Impact on User Engagement and Retention

Duolingo's gamification approach has yielded significant results:

- **Increased Retention Rates:**
 - Duolingo's internal data suggests that users who maintain a **7-day streak** are much more likely to continue learning long-term.
- **Higher Engagement:**
 - Studies, such as **Hamari et al. (2016)**, indicate that gamification leads to **higher user participation** in educational apps.
- **Personalized Learning Paths:**
 - **AI-driven adaptive difficulty** ensures users are neither underwhelmed nor overwhelmed, keeping them engaged.
- **Completion Rates:**
 - Compared to traditional e-learning platforms, Duolingo boasts **higher lesson completion rates**, thanks to its interactive and rewarding mechanics.

5. Challenges & Criticism of Duolingo's Gamification

Despite its success, Duolingo's approach faces several criticisms:

- **Superficial Learning:**
 - Critics argue that gamification encourages **rote memorization** rather than deep language comprehension.

- Sentence formation and conversational skills may be **neglected** in favor of repetitive drills.
- **Overemphasis on Competition:**
 - The leaderboard system may discourage users who feel **overwhelmed** by highly competitive peers.
 - Some learners may prioritize ranking over **actual language proficiency**.
- **Extrinsic vs. Intrinsic Motivation:**
 - Duolingo heavily relies on **external rewards (XP, badges)**, which may reduce **intrinsic motivation** for language learning over time.

6. Lessons for College ERP Gamification

Applying Duolingo's gamification model to a college ERP (**Enterprise Resource Planning**) system can increase student engagement:

- **Streaks and XP Systems:**
 - Encourages students to interact with the ERP daily (e.g., checking assignments, submitting forms on time).
- **Leaderboards:**
 - Can be used to motivate students to participate actively in discussions, complete assignments, and engage in academic activities.
- **Badges & Achievements:**
 - Rewards students for milestones such as **consistent attendance, assignment completion, and academic excellence**.
- **Personalized Challenges:**
 - AI-based difficulty adaptation can ensure students are challenged at an appropriate level, keeping them engaged and motivated.

7. Conclusion

- Duolingo's **gamification strategies** have **successfully increased engagement and retention** in language learning.
- By integrating reward-based learning, social motivation, and **behavioral reinforcement**, it creates an immersive and **habit-forming** experience.
- The principles of Duolingo's gamification model can be adapted to **various educational platforms, including college ERP systems**, to improve student participation and academic performance.



Key Gamification Features



Key Gamification Features

★ **XP Points & Levels** 🎯 – Earn points for attendance ✅, assignments 📄, and quizzes 🖋️.

🏆 **Badges & Achievements** 🌟 – Unlock badges for milestones (e.g., 100% attendance 📅).

📊 **Leaderboards** 🏆 – Foster healthy competition among peers.

🔥 **Streaks & Rewards** 📁 – Incentivize consistent engagement ⌚.

🕵️ **Quests & Challenges** 🚀 – Encourage participation in events & discussions 💬.



Prototype / Sample UI

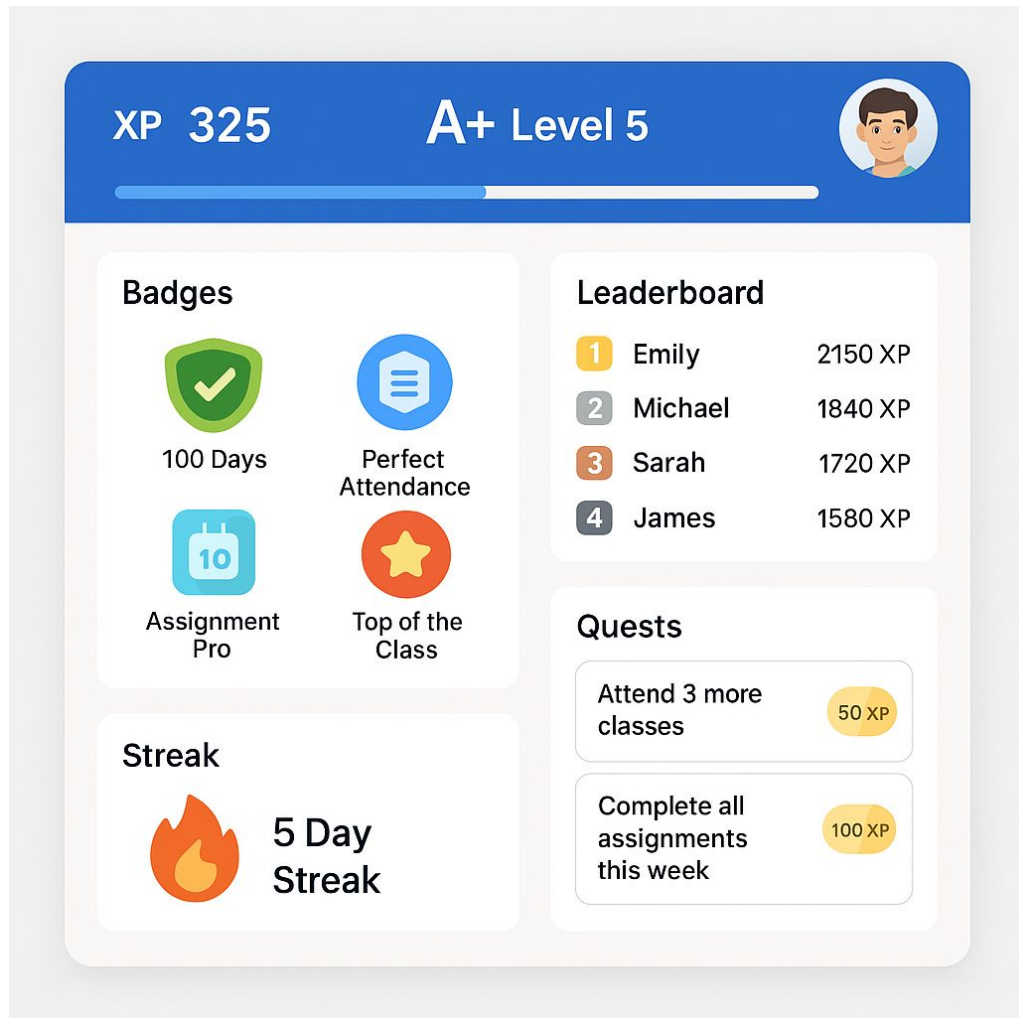
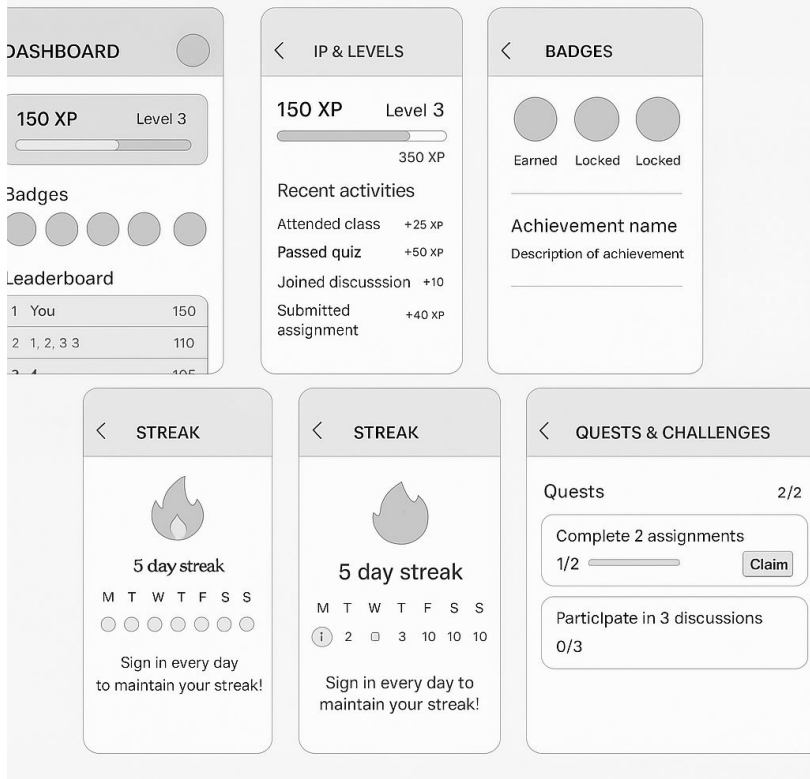




Prototype / Sample UI

Showcase a mockup of the gamified ERP dashboard.

Highlight how points, badges, and leaderboards are displayed.



Basic Brainstorming and Wireframing

Dashboard



CURRENT LEVEL

Level 7

2,450 XP / 3,000 XP to next level

[View Progress](#)



CURRENT STREAK

14 Days

Keep your streak going! Log in tomorrow.

[View Streak](#)



LEADERBOARD

Rank #12

You're in the top 15% this week!

[View Leaderboard](#)

Recent Badges



First Milestone

Completed your first 10 tasks

[View Details](#)



Team Player

Collaborated on 5 team projects

[View Details](#)



Quick Learner

Completed onboarding in record time

[View Details](#)



More Badges

Discover and unlock new achievements

[View All](#)

Active Quests



Complete Profile

Add a profile picture and bio (250 XP)



Daily Check-in

Log in for 5 consecutive days (100 XP)



Team Challenge

Collaborate with 3 team members (500 XP)



Leaderboard Preview

	 Sarah Johnson	Level 12	8,750 XP	
	 Michael Chen	Level 11	8,320 XP	
	 Jessica Williams	Level 10	7,890 XP	
	 You	Level 7	2,450 XP	

[View Full Leaderboard](#)



Badges & Achievements

All Badges Earned In Progress Locked

Your Progress



BADGE COMPLETION

12 of 36 Badges

You've earned 33% of all available badges

[View Statistics](#)

Recently Earned



EARNED TODAY

First Milestone

Completed your first 10 tasks

[Share](#)



EARNED 3 DAYS AGO

Team Player

Collaborated on 5 team projects

[Share](#)



EARNED 1 WEEK AGO

Quick Learner

Completed onboarding in record time

[Share](#)

In Progress



Consistency Champion

Log in for 30 consecutive days (70% complete)



Feedback Master

Provide feedback on 10 features (40% complete)



Social Butterfly

Connect with 20 team members (25% complete)



Badge Categories



Productivity

3 of 8 badges earned

[View Category](#)



Teamwork

4 of 10 badges earned

[View Category](#)



Learning

2 of 6 badges earned

[View Category](#)



Achievements

3 of 12 badges earned

[View Category](#)

Upcoming Badges



Innovation Star

Submit 3 improvement ideas (0/3 completed)



Perfect Attendance

Attend 10 team meetings on time (2/10 completed)



Knowledge Guru

Complete 5 training modules (1/5 completed)





Expected Outcomes



Expected Outcomes

 Increased ERP usage by students.

 Improved attendance  and assignment submission rates .

 Enhanced motivation through competition  and rewards .



Challenges and Considerations



Challenges and Considerations



Potential Challenges



Student Motivation Variability



Sustaining Engagement



Balancing Competition &
Collaboration



System Integration Complexity



Proposed Solutions



Adaptive Gamification



Long-term Engagement Strategies



Inclusive Design



Seamless ERP Integration



Future Scope



Future Scope



Personalization with AI – Customized challenges & rewards based on student behavior.



Expanding Gamification to Faculty – Encouraging teacher engagement through incentives.



Using Blockchain – Secure & transparent reward tracking 🏆.



Conclusion



Conclusion

Summary of Key Takeaways:

- ✓ Gamification is a proven method to boost engagement 🚀.
- 🏆 XP, badges, leaderboards, and streaks make ERP systems more interactive.
- 📖 ERP systems can evolve into engaging learning platforms 🎓, not just administrative tools.

Thank You!

Now, we're open for Questions.